

Incomplete Markets I

Bewley and Aiyagari Models

Macroeconomics — Lecture Notes

1. Complete vs Incomplete Markets

Complete Markets

A theoretical framework where all possible future risk can be hedged through trading securities for every state of nature. For every possible history s^t , there exists a corresponding Arrow security. Agents can achieve **perfect insurance** against any conceivable event.

Incomplete Markets

A more realistic scenario where only a limited number of securities exist. There are not enough financial instruments to cover all future risk. Agents cannot insure against every possible shock and instead rely on **self-insurance** through saving.

Bewley and Aiyagari models study economies with **incomplete markets**, where households face idiosyncratic risk but cannot perfectly insure against it.

2. Setup: Idiosyncratic Risk

2.1. States and Histories

- S_t : state of the world at time t — the realization of a stochastic event (e.g., employed or unemployed, high or low productivity).
- $s^t = \{s_1, s_2, \dots, s_t\}$: history of events up to time t , i.e., the complete sequence of realizations from the beginning until period t .
- $\pi(s^t)$: probability of history s^t occurring.

Note

S_t is a single realization at time t , while s^t is the full sequence up to t . For example, with states {sun, rain}, at $t = 2$ there are four possible histories: {sun, sun}, {sun, rain}, {rain, sun}, {rain, rain}. Each history has its own Arrow security and price.

2.2. Households

There is a continuum of infinitely lived households of measure 1, indexed $i \in [0, 1]$. All households have **rational expectations** — they forecast using the true probability distribution of the model.

Household i is endowed with $y_{i,t}(s^t)$ at history s^t . Aggregating across all households:

$$\int_0^1 y_{i,t}(s^t) di = Y_t(s^t)$$

where $Y_t(s^t)$ is the (random) **aggregate endowment**. By the law of large numbers, idiosyncratic shocks cancel in the aggregate, so Y_t reflects only aggregate risk.

2.3. Preferences and Aggregates

Households maximize their lifetime utility. Per-period utility is $u(c_{i,t}(s^t))$, where $c_{i,t}(s^t)$ denotes consumption of household i at history s^t . Aggregate consumption is:

$$\int_0^1 c_{i,t}(s^t) di = C_t(s^t)$$

Note

No intertemporal technology for now. There is no investment, no savings, and no storage. All resources produced in period t must be consumed in the same period:

$$C_t(s^t) = Y_t(s^t)$$

3. Two Benchmark Cases

Before studying incomplete markets, we analyze two extreme benchmarks.

	Autarky	Full Insurance
Markets	None	Complete
Insurance	None	Perfect
Income shock pass-through	100%	0%

3.1. Autarky

No markets exist. Each household consumes exactly its own endowment:

$$c_{i,t}(s^t) = y_{i,t}(s^t), \quad \forall i, s^t$$

This is **full pass-through**: if a household loses its job, its consumption falls one-for-one. There is no smoothing mechanism of any kind.

3.2. Full Insurance / Complete Markets

Markets are complete — an Arrow security exists for every possible history s^t . All contracts are negotiated at $t = 0$, before any uncertainty is resolved.

Arrow Securities

An Arrow security for history s^t pays **1 unit of consumption if s^t occurs, and 0 otherwise**. Its price at $t = 0$ is $p_t(s^t)$. With complete markets, a contract exists for every possible history, allowing perfect insurance against any event.

Time-Zero Arrow–Debreu Budget Constraint

Each household solves:

$$\max \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(c_{i,t}(s^t))$$

subject to the time-zero Arrow–Debreu budget constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} p_t(s^t) (c_{i,t}(s^t) - y_{i,t}(s^t)) = 0$$

The first summation runs over all periods $t = 0, 1, 2, \dots$; the second runs over all possible histories s^t within each period. The constraint says that the total net expenditure on Arrow securities, valued at prices $p_t(s^t)$, must equal zero: what the household spends on insurance for bad states is financed by selling insurance for good states.

4. Equilibrium under Full Insurance

4.1. First-Order Conditions and the Ratio of Marginal Utilities

The Lagrangian for household i is:

$$\mathcal{L} = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(c_{i,t}(s^t)) - \lambda_i \left[\sum_{t=0}^{\infty} \sum_{s^t} p_t(s^t) (c_{i,t}(s^t) - y_{i,t}(s^t)) \right]$$

where λ_i is the Lagrange multiplier on the lifetime budget constraint, measuring the **marginal value of wealth** for household i :

- A **rich** household has a low λ_i — an extra unit of wealth adds little utility.
- A **poor** household has a high λ_i — an extra unit of wealth adds a lot of utility.

The first-order condition with respect to $c_{i,t}(s^t)$ is:

$$\beta^t \pi(s^t) u'(c_{i,t}(s^t)) = \lambda_i p_t(s^t)$$

Writing the same condition for household j and dividing:

Key Result

$$\frac{u'(c_{i,t}(s^t))}{u'(c_{j,t}(s^t))} = \frac{\lambda_i}{\lambda_j}, \quad \forall t, s^t$$

The ratio of marginal utilities between any two households is **constant across all periods and states of the world**. The terms β^t , $\pi(s^t)$, and $p_t(s^t)$ cancel exactly.

This is a powerful implication: idiosyncratic income shocks are fully absorbed by the economy. If household i loses its job, society redistributes so that the relative consumption relationship between households does not change. The λ_i are **Pareto weights** — fixed at $t = 0$ by each household's initial wealth.

4.2. CRRA Utility and Proportionality to Aggregate Consumption

If utility is CRRA, $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, then $u'(c) = c^{-\sigma}$. Substituting into the ratio of marginal utilities:

$$\left(\frac{c_{i,t}(s^t)}{c_{j,t}(s^t)} \right)^{-\sigma} = \frac{\lambda_i}{\lambda_j} \implies \frac{c_{i,t}(s^t)}{c_{j,t}(s^t)} = \left(\frac{\lambda_i}{\lambda_j} \right)^{-\frac{1}{\sigma}}$$

The ratio of consumptions is also constant over time. Using the market clearing condition $\int_0^1 c_{i,t}(s^t) di = C_t(s^t)$, individual consumption can be expressed as:

Key Result

$$c_{i,t}(s^t) = \frac{\lambda_i^{-\frac{1}{\sigma}}}{\int_0^1 \lambda_j^{-\frac{1}{\sigma}} dj} C_t(s^t)$$

Individual consumption is **proportional to aggregate consumption**. The fraction $\frac{\lambda_i^{-1/\sigma}}{\int_0^1 \lambda_j^{-1/\sigma} dj}$ is a time-invariant share determined at $t = 0$ by relative wealth. If the economy grows, all households grow in the same proportion; if there is a recession, all fall in the same proportion. Idiosyncratic risk has completely disappeared — only aggregate risk remains.

where $\sigma > 0$ is the coefficient of relative risk aversion (CRRA parameter).

5. Partial Risk Sharing: Two Approaches

The Bewley/Aiyagari framework lies between the two benchmarks above. There are two approaches to modeling partial risk sharing:

5.1. 1. Exogenous Incomplete Markets

Simply **assume** that markets are incomplete and restrict the asset space to those assets observed in the real world (e.g., risk-free bonds, equities, mortgages). The incompleteness is imposed by the modeler — no explanation is given for why certain markets are missing. This is the approach followed in **Bewley and Aiyagari models**.

5.2. 2. Endogenous Incomplete Markets

The incompleteness arises endogenously from **fundamental frictions**:

- **Limited commitment:** Agents cannot commit to fulfilling future contracts. If tomorrow it is advantageous not to pay, they walk away. Because of this, no one offers full insurance in equilibrium.
- **Private information:** The insurer cannot observe the true situation of the insured. With full unemployment insurance, a household might not search for a job. This moral hazard problem implies that optimal insurance is incomplete.

End of Incomplete Markets I Notes